STAFF REVIEW · H446 057 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. New object's fields/properties.

2. Removes/returns oldest enqueued item.

3. Names/labels can repeat; ID enables unambiguous lookup.

4. 20, 21. The last row has index 2, so its values are 10*2+0 and 10*2+1.

I Do — source worked steps

1. Explain — Queue is FIFO, so the first enqueued request is processed first.

2. Why a file? — Unlike the queue, the text file can hold a record across program runs. openWrite creates a write stream and replaces any existing contents of the target file; it is not an append operation.

3. Trace the file contents — First writeLine stores Ada,Holes , then the second stores Ben,Dune . They become two separate text lines. This is a simple illustrative format, not a robust CSV parser.

4. Explain the effect — f.close() finishes the write operation and releases the file. Do not confuse storing records in a queue or dictionary with saving them permanently.

We Do — guided CHECK

1. Explain
A dictionary retrieves by key; it does not enforce FIFO service order.

2. Predict what is printed
Identify the role of endOfFile() and the effect of each readLine() call.
Follow two reads
The first call prints Ada,Holes ; the second prints Ben,Dune . The file position advances after each read.
Stop safely
Once no unread lines remain, endOfFile() is true, the loop stops, and f.close() releases the reader. This example reads text; it does not reconstruct Loan objects automatically.

You Do — source mark schemes

1. Related fields bundled [2]; object methods/encapsulation/maintainability [2].

2. Queue<Loan> q=new(); [2], two q.Enqueue(new Loan(...)) [4], if(q.Count>0) [1], Loan next=q.Dequeue() [1].

3. Dictionary<int,Loan> keyed by unique ID, reject duplicate keys [2]; validate a nonempty student name [2]; openWrite + writeLine (+ close) to create/write saved text [2]; openRead + readLine (+ close) to retrieve text [2].

Exit Ticket — source answer

1. False. Use a stable unique identifier.